

# DEEPANWITA BANERJEE

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## Research Interests

Systems Biology ♦ Metabolic Engineering ♦ Antibiotic resistance

## Skills

- **Metabolic modeling:** COBRA Toolbox, Genome scale metabolic reconstruction and analysis and computational strain optimization methods
- **Molecular Biology:** Cloning, PCR, Gene knockout using Lambda red
- NGS data analysis and primer designing for capillary electrophoresis sequencing
- Basic Biochemical, Microbiological techniques and Bioinformatics Tools
- Mammalian cell culture, MTT Assay, BIOLOG phenotypic microarray

## Patent

- Lee, T.S.; Wang, X.; Yunus, I.S.; Kim, J.; **Banerjee, D.**; Mukhopadhyay, A. Genetically modified *Pseudomonas* host cells and methods useful for producing isoprenol. **US Patent Application 18/976,085** (2025).
- **Eng, T.T.; Banerjee, D.; Mukhopadhyay, A.** Genetically modified bacterial cells and methods useful for producing indigoidine. **US Patent [11,767,521](#)** (2023).

## Latest Publications

*Total number of citations of 821 and google scholar h-index of 16. (\*Equal contributors)*

## Book Chapter

- **Bhattacharya, N.; Banerjee, D.**; O'Connor, C.; Parmar, D.; Moskovets, E.; *et al.* Direct Colony Atmospheric Pressure Matrix-Assisted Laser Desorption/Ionization Mass Spectrometry Assay for Targeted Operon Metabolite Analysis. In: **Metabolomics: Methods and Protocols**, 173–184 (2025).

## Peer Reviewed Journal Articles

1. Baral, N.R.; Yang, M.; **Banerjee, D.**; Mukhopadhyay, A; *et al.* Influence of titer, rate, yield, and scale on the cost and life-cycle emissions of biomanufacturing. **Trends in Biotechnology** (2026)
2. **Banerjee, D.**; Menasalvas, J.; Chen, Y.; Gin, J.W.; Baidoo, E.E.K.; Petzold, C.J.; *et al.* Addressing genome scale design tradeoffs in *Pseudomonas putida* for bioconversion of an aromatic carbon source. **npj Systems Biology and Applications** 11(1), 8 (2025).

3. Baral, N.R.; **Banerjee, D.**; Eng, T.; Simmons, B.A.; Mukhopadhyay, A.; Scown, C.D. Microbial pathways for cost-effective low-carbon renewable indigoidine. **ACS Sustainable Chemistry & Engineering** 13(8), 3300–3310 (2025).
4. Shrestha, S.; Goswami, S.; **Banerjee, D.**; Garcia, V.; Zhou, E.; Olmsted, C.N.; *et al.* Perspective on lignin conversion strategies that enable next generation biorefineries. **ChemSusChem** 17(16), e202301460 (2024).
5. Baral, N.R.; **Banerjee, D.**; Mukhopadhyay, A.; Simmons, B.A.; Singer, S.W.; *et al.* Integration of Genome-Scale Metabolic Model with Biorefinery Process Model Reveals Market-Competitive Carbon-Negative Sustainable Aviation Fuel Utilizing Microbial Cell Mass ... **BioResources** 19(3) (2024).
6. **Banerjee, D.\***; Yunus, I.S.\*; Wang, X.\*; Kim, J.\*; Srinivasan, A.; Menchavez, R.; *et al.* Genome-scale and pathway engineering for the sustainable aviation fuel precursor isoprenol production in *Pseudomonas putida*. **Metabolic Engineering** 82, 157–170 (2024).
7. Srinivasan, A.; Chen-Xiao, K.; **Banerjee, D.**; Oka, A.; Pidatala, V.R.; Eudes, A.; *et al.* Sustainable production of 2,3,5,6-Tetramethylpyrazine at high titer in engineered *Corynebacterium glutamicum*. **Journal of Industrial Microbiology & Biotechnology** 51, kuae026 (2024).
8. Baral, N.R.; **Banerjee, D.**; Mukhopadhyay, A.; Simmons, B.A.; Singer, S.W. Economic and environmental trade-offs of simultaneous sugar and lignin utilization for biobased fuels and chemicals. **ACS Sustainable Chemistry & Engineering** 12(7), 2563–2576 (2023).
9. Kulakowski, S.; **Banerjee, D.**; Scown, C.D.; Mukhopadhyay, A. Improving microbial bioproduction under low-oxygen conditions. **Current Opinion in Biotechnology** 84, 103016 (2023).
10. Eng, T.\*; **Banerjee, D.\***; Menasalvas, J.; Chen, Y.; Gin, J.; Choudhary, H.; Baidoo, E.; *et al.* Maximizing microbial bioproduction from sustainable carbon sources using iterative systems engineering. **Cell Reports** 42(9) (2023).
11. Garcia Martin, H.; Radivojevic, T.; Zucker, J.; Bouchard, K.; Sustarich, J.; Peisert, S.; *et al.* Perspectives for self-driving labs in synthetic biology. **Current Opinion in Biotechnology** 79, 102881 (2023).
12. **Banerjee, D.**; Mukhopadhyay, A. Perspectives in growth production trade-off in microbial bioproduction. **RSC Sustainability** 1(2), 224–233 (2023).

13. Czajka, J.J.; **Banerjee, D.**; Eng, T.; Menasalvas, J.; Yan, C.; Munoz, N.M.; *et al.* Tuning a high performing multiplexed-CRISPRi *Pseudomonas putida* strain to further enhance indigoidine production. **Metabolic Engineering Communications** 15, e00206 (2022).
14. **Banerjee, D.**; Eng, T.; Sasaki, Y.; Srinivasan, A.; Oka, A.; Herbert, R.A.; Trinh, J.; *et al.* Genomics Characterization of an Engineered *Corynebacterium glutamicum* in Bioreactor Cultivation Under Ionic Liquid Stress. **Frontiers in Bioengineering and Biotechnology** 9, 766674 (2021).
15. Lim, H.G.; Eng, T.; **Banerjee, D.**; Alarcon, G.; Lau, A.K.; Park, M.R.; Simmons, B.A.; *et al.* Generation of *Pseudomonas putida* KT2440 strains with efficient utilization of xylose and galactose via adaptive laboratory evolution. **ACS Sustainable Chemistry & Engineering** 9(34), 11512–11523 (2021).
16. Eng, T.\*; **Banerjee, D.\***; Lau, A.K.; Bowden, E.; Herbert, R.A.; Trinh, J.; Prahl, J.P.; *et al.* Engineering *Pseudomonas putida* for efficient aromatic conversion to bioproduct using high throughput screening in a bioreactor. **Metabolic Engineering** 66, 229–238 (2021).
17. **Banerjee, D.\***; Eng, T.\*; Lau, A.K.; Sasaki, Y.; Wang, B.; Chen, Y.; Prahl, J.P.; *et al.* Genome-scale metabolic rewiring improves titers, rates and yields of the non-native product indigoidine at scale. **Nature Communications** 11(1), 5385 (2020).
18. Wehrs, M.\*; Thompson, M.G.\*; **Banerjee, D.\***; Prahl, J.P.; Morella, N.M.; Barcelos, C.A.; *et al.* Investigation of Bar-seq as a method to study population dynamics of *Saccharomyces cerevisiae* deletion library during bioreactor cultivation. **Microbial Cell Factories** 19(1), 167 (2020).
19. Chen, Y.; **Banerjee, D.**; Mukhopadhyay, A.; Petzold, C.J. Systems and synthetic biology tools for advanced bioproduction hosts. **Current Opinion in Biotechnology** 64, 101–109 (2020).
20. **Banerjee, D.**; Raghunathan, A. Constraints-based analysis identifies NAD<sup>+</sup> recycling through metabolic reprogramming in antibiotic resistant *Chromobacterium violaceum*. **PLOS ONE** 14(1), e0210008 (2019).
21. Immanuel, S.R.C.\*; **Banerjee, D.\***; Rajankar, M.P.\*; Raghunathan, A. Integrated constraints based analysis of an engineered violacein pathway in *Escherichia coli*. **Biosystems** 171, 10–19 (2018).
22. **Banerjee, D.**; Raghunathan, A. Knowledge, attitude and practice of antibiotic use and antimicrobial resistance: a study post the ‘Red Line’ initiative. **Current Science** 114(9), 1866–1877 (2018).

23. **Banerjee, D.;** Parmar, D.; Bhattacharya, N.; Ghanate, A.D.; Panchagnula, V.; *et al.* A scalable metabolite supplementation strategy against antibiotic resistant pathogen *Chromobacterium violaceum* induced by NAD<sup>+</sup>/NADH<sup>+</sup> imbalance. **BMC Systems Biology** 11(1), 51 (2017).

## Preprints

- Baral, N. R.; Eng, T.; **Banerjee, D.;** Simmons, B. A.; Mukhopadhyay, A. and Scown, C. D. Cost and carbon implications of flexible biomanufacturing to produce bio-based gasoline, jet fuel, and platform chemicals. *Chemical Engineering Journal*, 177641, ISSN 1385-8947 (2026).

## Professional Experience

### BSE Division, Berkeley Lab

Emeryville, California, US

### Research Scientist, Mukhopadhyay Group

Aug 2021 – Present

- ◆ Evaluate microbial strain physiology during industrially relevant conditions.
- ◆ Implement hypotheses from metabolic models to generate complex engineered strains
- ◆ Examine wildtype and engineered microbial strains in a range of cultivation modes and regimes to analyze growth, production, and multi-omics data.

### Joint Bioenergy Institute – Berkeley Lab

Emeryville, California, US

### Postdoctoral Fellow, Mukhopadhyay Group

Oct 2018 – Aug 2021

- ◆ Developed a scalable model integrated workflow to predict strain engineering for improved production of bioproducts and biofuels in *Pseudomonas putida* KT2440.
- ◆ Predicted engineering strategies for growth coupled production of indigoidine in *P. putida* that were experimentally a success

### CSIR - National Chemical Laboratory

Pune, Maharashtra, India

### DST INSPIRE PhD Fellow, Advisor: Dr. Anu Raghunathan

Aug 2011 – May 2018

Dynamics of cell metabolism and enzymopathies in *Chromobacterium violaceum* – a systems approach.

- ◆ Identified genetic variations using next generation re-sequencing in genes including *acrR*, repressor of AcrAB drug efflux pump and S12 ribosomal protein RpsL
- ◆ Identified redox homeostasis is critical for survival using constraint-based modeling of core metabolism using COBRA Toolbox in MATLAB.
- ◆ Identified candidate metabolites that re-sensitize resistant populations taking advantage of redox imbalance by genome scale metabolic modeling of *C. violaceum*.

### National Institute of Technology Rourkela

Rourkela, Orissa, India

### M.Tech dissertation, Advisor: Dr. G. R. Satpathy

2008 - 2009

Red blood cell stabilization: effect of hydroxyethyl starch on RBC viability, functionality and oxidative state under different freeze thaw conditions.

- ◆ Isolation and Freeze drying of red blood cells.
- ◆ Protein and biochemical assays to identify oxidative stress

### Department Of Biotechnology, RANBAXY Laboratories

Gurugram, Haryana, India

**B. Tech dissertation, Supervisor: Dr. Roop Singh Borah** Jan – Jul 2007

- ◆ Gene cloning and Sub cloning and Mammalian cell culture
- ◆ Protein expression work (induction, SDS PAGE, ECL, Silver staining)

**Department of Biochemistry, All India Institute of Medical Sciences** Delhi, India

**Summer intern, Supervisor: Dr. Kalpana Luthra** July - Sep 2006

Resistance conferring genetic polymorphism of DC-SIGN, SDF-3'α in HIV-1

Seronegative high-risk individuals.

- ◆ RNA and DNA isolation, PCR and related techniques
- ◆ Genotyping: Gel electrophoresis for analyzing gene polymorphism
- ◆ Mammalian cell culture

## **Mentoring & Teaching Experience**

**Joint Bioenergy Institute – Berkeley Lab**

Emeryville, California, US

**Postdoc, Mukhopadhyay Group.**

Oct 2018 – Aug 2021

- ◆ Mentored a 2021 SULI summer research intern for the successful completion of the project titled “Genome Scale Metabolic Model of a Microbial Host, *Pseudomonas putida*”
- ◆ Mentored a 2020 SULI summer research intern for the successful completion of the project titled “Predicting Engineering Targets in *Corynebacterium glutamicum* to Increase Production of Indigoidine”

**CSIR – National Chemical Laboratory**

Pune, Maharashtra, India

**Phd, Advisor: Dr. Anu Raghunathan**

Aug 2011 – May 2018

- ◆ Active management and mentorship of undergraduate and masters lab interns
- ◆ Reviewed research articles for peer reviewed Journals
- ◆ Co-manager for a 2-day meeting on Mathematical Modeling of National and Synthetic Genetic Networks, March 2017, Pune, India.
- ◆ Lead manager and organizer for a 3-day Metabolic Modeling Workshop, MetaMod I, March 2016, CSIR – NCL and IISER Pune, India.

**Lovely Professional University**

Punjab, India

**Assistant Professor – Department of Biotechnology**

Jul 2010 – Jul 2011

- ◆ **Subjects** - Fuel Biotechnology, Immunology, Advances in Microbiology, Animal Cell Technology and Stem Cell Technology
- ◆ Supervised 8 undergraduate (B.Tech) and 2 masters (M.Tech) students

## **Invited Talks**

- ◆ “Engineering *Pseudomonas putida* for efficient conversion of lignin derived sugars and aromatics into bioproducts and biofuels” in SIMB 71<sup>st</sup> Annual Meeting, held at Austin Texas, August 10, 2021.
- ◆ “Genome-scale Metabolic Rewiring for Engineering Robust Production Platforms” in Indo-US workshop on “Recent Developments in Bioenergy Research”, held at IIT Kharagpur, October 19, 2020.

## **Conference Presentations**

- ◆ **Banerjee, D.**, Eng, T., Lau, A. K., Sasaki, Y., Herbert, R.A. Chen, Y., ..... Keasling, J. D., Mukhopadhyay, A. Strain Engineering in *Pseudomonas putida* Using High Throughput Fitness Profiling and Genome Scale Metabolic Models. Poster presentation at Metabolic Engineering 14 Conference, July 11 – 15, 2021.
- ◆ **Banerjee, D.**, Awasthi, D., Lim, H. G., Diaz, R. J., Eng, T., Simmons, B. A., Singer, S. W., Feist, A., Mukhopadhyay, A. and Keasling J. D. Integrated *Pseudomonas Putida* Strain Design for Maximizing Biomass Conversion to Biofuels and Bioproducts. Poster presentation at US DOE Office of Biological and Environmental Research (BER), 2021 Genomic Sciences Program (GSP) Annual PI Meeting, February 22 – 24, 2021.
- ◆ **Banerjee, D.**, Lau, A., Wang, B., Prahl JP., Tanjore, D. Eng, T., and Mukhopadhyay, A. Systems Metabolic Engineering of *Pseudomonas putida* for indigoidine production. Poster presentation at JBEI Annual DOE Review, December 3 – 4, 2019
- ◆ **Banerjee, D.** Knowledge Awareness and Practice of AMR Post Redline Initiative. Invited talk at a Meeting on Antibiotic Resistance at IISER Pune, October 28, 2017.
- ◆ **Banerjee, D.**, Bhattacharya, N., Panchagnula, V. and Raghunathan, A. Integrating metabolomic data as constraints in flux balance models to understand antibiotic resistance. Poster presentation delivered at 11th Annual International Conference, Metabolomics 2015, CA, USA.

## Awards and Honors

- ◆ 2026 Processes MDPI Journal Reviewer Board Member
- ◆ More than 6 years' experience as Reviewer for Journals including Current Microbiology, Metabolites, Scientific Reports, Nature Communication, etc.
- ◆ 2021 JBEI annual award in Research contribution
- ◆ 2020 SPOT recognition award by Berkeley Lab.
- ◆ 2020 JBEI annual award in Outreach and Community contribution.
- ◆ 2015 DBT-CTEP Travel Grant
- ◆ 2011 DST INSPIRE Ph.D. Fellowship
- ◆ 2009-2010 BITP training sponsored by Department of Biotechnology (DBT), Government of India
- ◆ 2009 First Rank in M.Tech Biotechnology, Institute Silver Medal
- ◆ 2007 MHRD scholarship for M.Tech Biotechnology
- ◆ 2007 GATE, Life Sciences

## Education

**CSIR - National Chemical Laboratory**

Ph. D., Biological Sciences

Pune, Maharashtra, India

June 2018

**National Institute of Technology**

M.Tech. Biotechnology

Rourkela, Orissa, India

2009

**Maharshi Dayanand University**

B. Tech. Biotechnology

Rohtak, Haryana, India

2007